Comparative-Static Analysis of General-Function Models

ECON 320 - Introduction to Mathematical Economics

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Contents

1 Differentials

For a single-variable function $y = f(x)$, a *small* change Δx induces a change in y that is well approximated by the *differential*

$$\mathrm{d}y \;=\; f'(x)\,\mathrm{d}x.$$

Formally, $\Delta y = f'(x)\Delta x + o(\Delta x)$ where $o(\Delta x)/\Delta x \to 0$ as $\Delta x \to 0$.

Linear approximation & error intuition

The tangent line at $(x_0, f(x_0))$ is

$$L(x) \;=\; f(x_0) + f'(x_0)(x - x_0),$$

so for small Δx, $f(x_0 + \Delta x) \approx L(x_0 + \Delta x)$ and the approximation error is second order.

Point elasticity via differentials

If $y = f(x) > 0$ and $x > 0$, the (arc-)point elasticity is

$$\varepsilon_{y,x} \;\equiv\; \frac{\mathrm{d}y}{\mathrm{d}x}\cdot\frac{x}{y} \;=\; \frac{\mathrm{d}\ln y}{\mathrm{d}\ln x}.$$

Quick use: $\mathrm{d}\ln y \approx \varepsilon_{y,x}\,\mathrm{d}\ln x$. If quantity $Q(p)$ has price elasticity $\varepsilon_{Q,p}$, then a 1% change in p yields about a $\varepsilon_{Q,p}$% change in Q.

Example 8.1 (Elasticity from a log form). If $\ln y = \alpha + \beta \ln x$, then $\varepsilon_{y,x} = \beta$ at every point. A 5% increase in x raises y by roughly 5β%.

2 Total Differentials

Definition

For $z = f(x_1, \ldots, x_n)$ with f differentiable,

$$\mathrm{d}z \;=\; \sum_{i=1}^{n} f_{x_i}\,\mathrm{d}x_i \;=\; f_{x_1}\,\mathrm{d}x_1 + \cdots + f_{x_n}\,\mathrm{d}x_n.$$

This is the best linear approximation to Δz from small simultaneous changes in the arguments.

Example 8.2 (Cobb–Douglas production). Let $Q = f(K, L) = AK^{\alpha}L^{1-\alpha}$ with $A > 0$, $\alpha \in (0, 1)$. Then

$$\mathrm{d}Q \;=\; \alpha A K^{\alpha-1}L^{1-\alpha}\,\mathrm{d}K \;+\; (1-\alpha)AK^{\alpha}L^{-\alpha}\,\mathrm{d}L.$$

In growth-rate form: $\mathrm{d}\ln Q = \alpha\,\mathrm{d}\ln K + (1-\alpha)\,\mathrm{d}\ln L$.

Example 8.3 (Price index). For $P = \left(\sum_i w_i p_i^{\theta}\right)^{1/\theta}$, the total differential is

$$\mathrm{d}P = \frac{1}{\theta}\left(\sum_i w_i p_i^{\theta}\right)^{\frac{1}{\theta}-1}\left[\sum_i w_i \theta p_i^{\theta-1}\,\mathrm{d}p_i\right] = \sum_i \omega_i\,\mathrm{d}p_i,$$

with weights $\omega_i = \dfrac{w_i p_i^{\theta-1}}{\sum_j w_j p_j^{\theta-1}}$.

3 Rules of Differentials

For differentiable $u, v > 0$ and constant a:

(i) $\mathrm{d}(a) = 0, \quad \mathrm{d}(au) = a\ \mathrm{d}u.$

(ii) **Sum/difference:** $\mathrm{d}(u \pm v) = \mathrm{d}u \pm \mathrm{d}v.$

(iii) **Product:** $\mathrm{d}(uv) = u\ \mathrm{d}v + v\ \mathrm{d}u.$

(iv) **Quotient:** $\mathrm{d}\left(\frac{u}{v}\right) = \frac{v\ \mathrm{d}u - u\ \mathrm{d}v}{v^2}.$

(v) **Power:** $\mathrm{d}(u^n) = nu^{n-1}\,\mathrm{d}u$ for any real n.

(vi) **Exponential/log:** $\mathrm{d}(e^u) = e^u\ \mathrm{d}u, \quad \mathrm{d}(\ln u) = \frac{\mathrm{d}u}{u}.$

Example 8.4 (Average–marginal link). If $AC(Q) = C(Q)/Q$, then

$$\mathrm{d}AC = \frac{Q\ \mathrm{d}C - C\ \mathrm{d}Q}{Q^2} \;\Rightarrow\; \left.\frac{\mathrm{d}AC}{\mathrm{d}Q}\right|_Q = \frac{MC - AC}{Q}.$$

AC rises when $MC > AC$ and falls when $MC < AC$.

4 Total Derivatives

Chain rule with one endogenous channel

Suppose $z = f(x, y)$ and $y = y(x)$. A small change in x induces

$$\frac{\mathrm{d}z}{\mathrm{d}x} \;=\; f_x + f_y\,\frac{\mathrm{d}y}{\mathrm{d}x}.$$

This is the total derivative—it captures both the direct effect (f_x) and the *indirect* effect through y.

Example 8.5 (Profit with demand feedback). Let $\pi(p) = p\,Q(p) - C(Q(p))$. Then

$$\frac{\mathrm{d}\pi}{\mathrm{d}p} = Q(p) + pQ'(p) - C'(Q)Q'(p) = Q + \underbrace{(p - MR(Q))\,Q'}_{\text{feedback term}},$$

where $MR(Q) \equiv C'(Q)$ at optimum. The term $pQ'(p)$ appears because Q responds to p.

Chain rule with multiple channels

If $z = f(\boldsymbol{x}; \alpha)$, $x_i = x_i(\alpha)$,

$$\frac{\mathrm{d}z}{\mathrm{d}\alpha} = f_\alpha + \sum_i f_{x_i}\,\frac{\mathrm{d}x_i}{\mathrm{d}\alpha}.$$

Variation on a theme (Parameters). Let $U(x, y; \theta)$ be an indirect utility with a taste parameter θ. Then

$$\frac{\mathrm{d}U}{\mathrm{d}\theta} = U_\theta + U_x x_\theta + U_y y_\theta.$$

At a Marshallian optimum, envelope logic often simplifies U_x and U_y terms (see Varian, *Microeconomic Analysis*, ch. 5).

5 Derivatives of Implicit Functions

One equation, one endogenous variable

If $F(x, y) = 0$ defines $y = y(x)$ and $F_y \neq 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} \;=\; -\frac{F_x}{F_y}.$$

Derivation: Totally differentiate $F(x, y) = 0$:

$$F_x \;\mathrm{d}x + F_y \;\mathrm{d}y = 0 \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y}.$$

Example 8.6 (Keynesian income with general consumption). Equilibrium: $Y = C(Y) + I + \bar{G}$. Define $F(Y; \bar{G}) \equiv Y - C(Y) - I - \bar{G} = 0$. Then

$$\frac{\mathrm{d}Y}{\mathrm{d}\bar{G}} = -\frac{F_{\bar{G}}}{F_Y} = \frac{1}{1 - C'(Y)} \;>\; 0 \quad \text{if } C'(Y) \in (0, 1).$$

This reproduces the multiplier without solving the reduced form explicitly.

Multivariate implicit functions via matrices

Let $\boldsymbol{F}(\boldsymbol{x}; \alpha) = \mathbf{0}$ with $\boldsymbol{x} = (x_1, \ldots, x_n)'$. The $n \times n$ Jacobian

$$J \;\equiv\; \frac{\partial \boldsymbol{F}}{\partial \boldsymbol{x}} \;=\; \begin{bmatrix} F_{1x_1} & \cdots & F_{1x_n} \\ \vdots & \ddots & \vdots \\ F_{nx_1} & \cdots & F_{nx_n} \end{bmatrix}.$$

Differentiating w.r.t. parameter α:

$$J\,\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\alpha} + \frac{\partial \boldsymbol{F}}{\partial \alpha} = \mathbf{0} \;\;\Rightarrow\;\; \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\alpha} \;=\; -\,J^{-1}\frac{\partial \boldsymbol{F}}{\partial \alpha},$$

provided $\det J \neq 0$ (implicit-function conditions).

Cramer's rule representation (qualitative signs). Let $G \equiv \frac{\partial \boldsymbol{F}}{\partial \alpha}$. Then the ith component is

$$\frac{\mathrm{d}x_i}{\mathrm{d}\alpha} = -\,\frac{\det J^{(i)}}{\det J},$$

where $J^{(i)}$ replaces the ith column of J by G. This is useful for sign analysis when only partial-derivative signs are known.

6 Comparative Statics of General-Function Models

Why "general-function"?

Often economic models are given by *behavioral* and *equilibrium* equations that do not admit convenient explicit (reduced-form) solutions. Comparative statics proceeds by total differentiation of the defining system.

Two-equation market model

Let price p and quantity Q be endogenous; income m and input price w exogenous. With general demand and supply,

$$\begin{cases} F_1(p,Q;m) \equiv Q - D(p,m) = 0, \\ F_2(p,Q;w) \equiv Q - S(p,w) = 0. \end{cases}$$

The Jacobian and parameter blocks are

$$J = \begin{bmatrix} -D_p & 1 \\ -S_p & 1 \end{bmatrix}, \qquad \frac{\partial \boldsymbol{F}}{\partial m} = \begin{bmatrix} -D_m \\ 0 \end{bmatrix}.$$

Solve $J \begin{bmatrix} \mathrm{d}p \\ \mathrm{d}Q \end{bmatrix} + \begin{bmatrix} -D_m \\ 0 \end{bmatrix} \mathrm{d}m = \mathbf{0}$, so $\begin{bmatrix} \mathrm{d}p \\ \mathrm{d}Q \end{bmatrix} = J^{-1} \begin{bmatrix} D_m \\ 0 \end{bmatrix} \mathrm{d}m$. By Cramer,

$$\frac{\mathrm{d}p}{\mathrm{d}m} = \frac{\det \begin{bmatrix} D_m & 1 \\ 0 & 1 \end{bmatrix}}{\det J} = \frac{D_m}{S_p - D_p}, \qquad \frac{\mathrm{d}Q}{\mathrm{d}m} = \frac{\det \begin{bmatrix} -D_p & D_m \\ -S_p & 0 \end{bmatrix}}{\det J} = \frac{D_m S_p}{S_p - D_p}.$$

With $D_p < 0$, $S_p > 0$, $D_m > 0$ we have $\mathrm{d}p/\mathrm{d}m > 0$ and $\mathrm{d}Q/\mathrm{d}m > 0$.

Interpretation. Income shifts demand out. Whether p or Q moves more depends on the *relative slopes* D_p and S_p; steeper supply (large S_p) tilts the adjustment toward price changes.

Tax incidence in the general model (per-unit tax t)

Let the consumer pays p, firm receives $p - t$:

$$\begin{cases} Q - D(p,m) = 0, \\ Q - S(p - t, w) = 0. \end{cases}$$

Differentiate w.r.t. t. The parameter block is $G = (0,\ +S_p)'$. With $\det J = S_p - D_p > 0$,

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{S_p}{S_p - D_p}, \qquad \frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{-\,D_p S_p}{S_p - D_p}.$$

Incidence: $0 < \mathrm{d}p/\mathrm{d}t < 1$. The side with the *less elastic* curve bears more of the tax (Varian, ch. 2).

Keynesian income model without solving the reduced form

Equilibrium: $Y = C(Y - T) + I + \bar{G}$ with T exogenous for simplicity. Define $F(Y; \bar{G}) \equiv Y - C(Y - T) - I - \bar{G} = 0$. Then

$$\frac{\mathrm{d}Y}{\mathrm{d}\bar{G}} = \frac{1}{1 - C'(Y - T)} = \frac{1}{1 - c} \quad \text{when } C'(Y - T) = c.$$

The method works even when $C(\cdot)$ is nonlinear, where the usual "solve-for-Y" trick is messy.

General $n \times n$ comparative statics recipe

1. Write the system $\boldsymbol{F}(\boldsymbol{x};\alpha) = \boldsymbol{0}$ and identify endogenous $\boldsymbol{x}$ and exogenous α.
2. Compute the Jacobian $J = \frac{\partial \boldsymbol{F}}{\partial \boldsymbol{x}}$ and check $\det J \neq 0$ (existence/uniqueness, IFT).
3. Totally differentiate: $J\,\mathrm{d}\boldsymbol{x} + \frac{\partial \boldsymbol{F}}{\partial \alpha}\,\mathrm{d}\alpha = \boldsymbol{0}$.
4. Solve: $\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\alpha} = -J^{-1}\frac{\partial \boldsymbol{F}}{\partial \alpha}$ (Cramer's rule for signs).
5. **Qualitative signs:** Use known signs of partials to infer signs of minors and hence the signs of the effects.

Worked 2×2 template (memorize). For

$$\begin{cases} F_1(x,y;\alpha) = 0, \\ F_2(x,y;\alpha) = 0, \end{cases} \qquad J = \begin{bmatrix} F_{1x} & F_{1y} \\ F_{2x} & F_{2y} \end{bmatrix},\ G = \begin{bmatrix} F_{1\alpha} \\ F_{2\alpha} \end{bmatrix},$$

$$\frac{\mathrm{d}x}{\mathrm{d}\alpha} = -\frac{\det \begin{bmatrix} F_{1\alpha} & F_{1y} \\ F_{2\alpha} & F_{2y} \end{bmatrix}}{\det J}, \qquad \frac{\mathrm{d}y}{\mathrm{d}\alpha} = -\frac{\det \begin{bmatrix} F_{1x} & F_{1\alpha} \\ F_{2x} & F_{2\alpha} \end{bmatrix}}{\det J}.$$

Mnemonic: replace the column of the variable you want by G, take the determinant, divide by $\det J$, and stick on a minus sign.

7 Limitations of Comparative Statics

- **Local/linear:** Results are first-order (tangent-line) approximations. Large parameter changes may require higher-order terms or global analysis.
- **Equilibrium selection:** If multiple equilibria exist, comparative statics describes *one* local branch around a point. Dynamics or stability are not addressed here.
- **Sign indeterminacy:** Without information on partial-derivative *magnitudes* (or the signs of cofactors), directions of change can be ambiguous.
- **Regularity conditions:** Differentiability and a nonsingular Jacobian are essential (implicit-function theorem). Kinks, corners, and discrete adjustments are outside the scope.
- **Path vs. endpoints:** Comparative statics compares equilibria—it says *nothing* about the time path (adjustment dynamics are in later chapters).

Extended worked examples

E1. Nested feedback in a general demand system

Two goods with demands $Q_1 = D_1(p_1,p_2,m)$, $Q_2 = D_2(p_1,p_2,m)$; market clearing fixes (p_1,p_2) given $(m,\bar{Q}_1,\bar{Q}_2)$:

$$\begin{cases} \bar{Q}_1 - D_1(p_1,p_2,m) = 0, \\ \bar{Q}_2 - D_2(p_1,p_2,m) = 0. \end{cases}$$

Let $\alpha = m$. The Jacobian is $J = -\begin{bmatrix} D_{1p_1} & D_{1p_2} \\ D_{2p_1} & D_{2p_2} \end{bmatrix}$ and $G = -\begin{bmatrix} D_{1m} \\ D_{2m} \end{bmatrix}$. Then

$$\frac{\mathrm{d}p_1}{\mathrm{d}m} = -\frac{\det \begin{bmatrix} -D_{1m} & -D_{1p_2} \\ -D_{2m} & -D_{2p_2} \end{bmatrix}}{\det J} = \frac{\det \begin{bmatrix} D_{1m} & D_{1p_2} \\ D_{2m} & D_{2p_2} \end{bmatrix}}{\det \begin{bmatrix} D_{1p_1} & D_{1p_2} \\ D_{2p_1} & D_{2p_2} \end{bmatrix}}.$$

Cross-price effects D_{ip_j} govern whether income raises p_1 more than p_2.

E2. Imperfect competition with cost shifter

A monopolist faces $Q = D(p)$, cost $C(Q;\omega)$ with cost shifter ω (e.g. wage). FOC: $MR(Q) = MC(Q;\omega)$. Write $F(Q;\omega) \equiv p(Q) + Qp'(Q) - C_Q(Q;\omega) = 0$.

$$\frac{\mathrm{d}Q^*}{\mathrm{d}\omega} = -\frac{F_\omega}{F_Q} = \frac{C_{Q\omega}}{2p'(Q) + Qp''(Q) - C_{QQ}(Q;\omega)}.$$

Under $p' < 0$, $p'' \le 0$, $C_{QQ} \ge 0$, the denominator is < 0 and $C_{Q\omega} > 0$ implies $\mathrm{d}Q^*/\mathrm{d}\omega < 0$.

E3. Comparative statics in IS–LM (qualitative)

Let Y, r be endogenous; government spending G a parameter:

$$\begin{cases} F_1(Y, r; G) \equiv Y - C(Y - T) - I(r) - G = 0, \\ F_2(Y, r; M) \equiv M/P - L(Y, r) = 0. \end{cases}$$

$J = \begin{bmatrix} 1 - C_Y & I_r \\ -L_Y & -L_r \end{bmatrix}$, $G = (-1,\, 0)'$. Then

$$\frac{\mathrm{d}Y}{\mathrm{d}G} = -\frac{\det \begin{bmatrix} -1 & I_r \\ 0 & -L_r \end{bmatrix}}{\det J} = \frac{L_r}{(1 - C_Y)L_r + I_r L_Y} \;>\; 0,$$

given $C_Y \in (0,1)$, $I_r < 0$, $L_Y > 0$, $L_r > 0$. The interest-rate response is

$$\frac{\mathrm{d}r}{\mathrm{d}G} = -\frac{\det \begin{bmatrix} 1 - C_Y & -1 \\ -L_Y & 0 \end{bmatrix}}{\det J} = \frac{L_Y}{(1 - C_Y)L_r + I_r L_Y} \;>\; 0.$$

Study guide & tips

- **Always label variables by role.** Pick *endogenous* vector $\boldsymbol{x}$ and *exogenous* parameters α early; it clarifies J and G immediately.
- **Check the Jacobian sign.** For 2×2 systems, $\det J > 0$ is a common regularity condition ensuring a unique local solution.
- **Use log-differentials for elasticities.** They turn percentage changes into additive terms and make multiplicative functions linear.
- **Cramer's rule for signs.** Even without solving, you can learn a lot by replacing one column and inspecting signs of the resulting 2×2 determinant.

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